LESSON PLAN

ADVANCED AIRCRAFT TECHNOLOGY (20)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	12
Total No. of Lesson Plans	20	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Aero engine classical control system
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Aero engine classical control system.
5. Specific Objectives	20.4.2.1 Trainee will be able to explain principles of Aero engine control systems.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of PPT diagrams; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of an aero engine control system?
(b) Name two types of engine control systems used in aviation.
(c) Why is the classical control system still relevant in modern training?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of Classical Control System • Throttle Lever Operation Principles • Governor Mechanism Functionality • Fuel Flow Regulation Basics • Hydraulic Actuation Systems	L & Ds	PPT, IB	15 min
(b)	• RPM Control via Governor • Load Sensing and Response • Mechanical Linkage Geometry • Safety Valves and Limits • Idle and Takeoff Settings	L & Ds	PPT, PDF	20 min
(c)	• Comparison with Fly-by-Wire • Advantages of Mechanical Systems • Limitations of Classical Design • Maintenance Requirements • Transition to Electronic Control	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the governor maintains constant RPM during load changes.
(b) Describe the role of the mechanical linkage in throttle control.
(c) List three advantages of classical control systems over fly-by-wire.

12. Home Assignment and Follow Up	Prepare a brief note comparing the mechanical feedback loop of the classical system with the electronic feedback loop of the Fly-by-Wire system to be discussed in LP 13.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Explanation of Fly by wire control systems using Fly by Wire demonstrator at Control Lab.	Refer to the master syllabus reference list	LP 13

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min